Laurence Datrier

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EDUCATION

PHD IN PHYSICS AND ASTRONOMY University of Glasgow, Institute for Gravitational Research
Glasgow, UK | October 2017 - June 2023
"Exploring the calibration of cosmological probes used in gravitational-wave and multi-messenger astronomy",
supervised by Prof. Martin Hendry and Prof. Graham Woan.
Funded by STFC as part of ScotDist, the Scottish Data-Intensive Science Triangle.

MSC BY RESEARCH IN PHYSICS AND ASTRONOMY University of Glasgow, Institute for Gravitational Research
Glasgow, UK | October 2016 - June 2018
"Investigating the prospects for constraining the Hubble constant using compact binary coalescences as standard sirens",
supervised by Prof. Martin Hendry and Dr. Chris Messenger.

BSC IN PHYSICS AND ASTRONOMY University of Glasgow
Glasgow, UK | September 2011 - June 2016
Final grade awarded: 2:1
Extra-curricular: Astronomy Society President, Physics Society Publicity Officer 2015-2016

BA IN FILM AND TV STUDIES Brunel University
Uxbridge, UK | September 2008 - June 2011

WORK

POSTDOCTORAL ASSOCIATE California State University Fullerton
Fullerton, CA, USA | February 2025 - Current
Postdoctoral Associate in Gravitational-Wave Physics for the Cosmic Explorer Project, as part of the Site Evaluation team.

HONORARY RESEARCHER University of Glasgow
Glasgow, UK | August 2023 - Current
Gravitational wave cosmology and multi-messenger astronomy work

PLANETARIUM PRESENTER Glasgow Science Centre
Glasgow, UK | January 2023 - November 2023
Full time from January to March 2023, casual worker from March 2023.

- Presenting planetarium shows to all audiences
- Helping with the running of daily operations at the GSC

SUMMER INTERN CAMK PAN Torun
Torun, Poland | August 2015
Model fitting of stellar evolution models and chemical abundance data for red giants in stellar clusters, using MESA and python

ASTRONOMY OUTREACH PROGRAMME DEMONSTRATOR University of Glasgow
Glasgow, UK | March 2015 - 2021
Outreach during March 2015 solar eclipse viewing event • Multiple tours of the Acre Road Observatory for various audiences • Astrophysics workshops for high school students • Stargazing nights • Science Festivals

VISITS AND SUMMER SCHOOLS

VISITING SCIENTIST Caltech
Pasadena, CA, USA | August 2019
Mentor: Michael Coughlin
Three week visit at Caltech to discuss projects involving the application of Gaussian Process Regression to kilonova model interpolation.

LSC FELLOW LIGO Hanford
Richland, WA, USA | May 2019 - August 2019
Mentors: Rick Savage, Sheila Dwyer, Jenne Driggers

- Data analysis for the photon calibrator, investigation of temperature dependence of integrating spheres, other routine measurements
- Data acquisition and modelling for the Newtonian Calibrator
- Investigation of the impact of wind gusts on detector duty cycles and observing range, in the context of building a wind fence around the end stations
- Work on offline feedforward filters

PHYSICS OF MACRONOVAE SUMMER SCHOOL Stockholm, Sweden | June 2018

CONFERENCES AND TALKS

AAPT SUMMER MEETING 2026 Invited Talk
Pasadena, CA, USA | July 2026
"Inspiring the future generation of gravitational wave scientists with Cosmic Explorer"

CALIFORNIA STATE UNIVERSITY, FULLERTON Invited Colloquium
Fullerton, CA, USA | May 2026
"Exploring the gravitational wave cosmos"

INSTITUTE FOR GRAVITATIONAL RESEARCH MARCH 2026 MEETING Talk
Glasgow, UK | March 2026
"Update on Cosmic Explorer site evaluation"

APS MEETING MARCH 2026 Talk
Denver, CO, USA | March 2026
"Site evaluation and cost estimation for Cosmic Explorer"

LVK MEETING SEPTEMBER 2025 Poster
Fort Collins, CO, USA | September 2025
"Site evaluation and cost estimation for Cosmic Explorer"

24TH INTERNATIONAL CONFERENCE ON GENERAL RELATIVITY AND GRAVITATION (GR24) AND 16TH EDOARDO AMALDI CONFERENCE Talk
Glasgow, UK | July 2025
"Site evaluation and cost estimation for Cosmic Explorer"

NAM 2022 Poster
Warwick, UK | July 2022
"A robust test of galaxy catalogue completeness for cosmology with dark standard sirens"

CIERA NORTHWESTERN Invited Virtual Talk
Evanston, IL, USA | October 2021
"A robust test of galaxy catalogue completeness for cosmology with dark standard sirens"

NAM 2021 Talk
Online conference, UK | July 2021
"Prospects for multi-messenger astronomy with serendipitous kilonova discoveries with LSST"

LIGO HANFORD Invited Talk
International Physics and Astronomy Educators Program
LIGO Hanford, WA, USA | June 2019
"Kilonovae, Neutron Stars and Gravitational Waves"

MERGING VISIONS
Exploring Compact-Object Binaries with Gravity and Light
Santa Barbara, WA, USA | June 2019

CORMACK MEETING Poster
Edinburgh, UK | November 2018
"Constraining binary neutron star merger times from kilonova observations"

UCL MACHINE LEARNING CONFERENCE Poster
London, UK | July 2018
"Detecting and classifying kilonovae"

7 MINUTES OF SCIENCE Public Talk
Glasgow, UK | February 2018
Public talk on gravitational wave cosmology

CAFE SCIENTIFIQUE EN FRANCAIS Invited Public Talk
Alliance francaise, Glasgow, UK | February 2017
Public talk on the detection of gravitational waves, in French

CONFERENCE FOR ASTRONOMY AND PHYSICS STUDENTS Organiser, Public Talk
Glasgow, UK | June 2016
Conference Organiser • Public talk: "The Physics of Interstellar"

SERVICE

PHYSICAL REVIEW LETTERS
2026 | One review

CLASSICAL AND QUANTUM GRAVITY
2026 | One review

OUTREACH

I'M A SCIENTIST, GET ME OUT OF HERE
Summer 2019
Online chats with students in the UK and other countries

LIGO HANFORD TOUR GUIDE
LIGO Hanford, WA, USA | May 2019 - August 2019
Leading tours of the LIGO Hanford Observatory for the public

JIMFEST
Glasgow, UK | February 2019
Volunteer, brochure design for the celebration of Sir James Hough's work

FESTIVAL OF PHYSICS
Glasgow, UK | February 2018
Volunteer for Institute for Gravitational Research stall

PRIZES AND AWARDS

RESEARCH FIRSTS EXHIBITION
Winning Entry
June 2023
The Triumph of GW150914 - Artwork

LIGO LABORATORY AWARD FOR EXCELLENCE IN DETECTOR CHARACTERIZATION AND CALIBRATION
Honorable mention
2020
"For the development, deployment, and commissioning of a novel absolute calibration system via direct gravitational forces, a 'Newtonian Calibrator', to provide increased confidence in the absolute calibration of LIGO detector data in future observing runs."

TANNAHILL BEQUEST
2015-2016
Best Astronomy 4H Student

ASTRONOMY 4H PRIZE
2015-2016
Best Astronomy 4H Lab Student

KERR BURSARY
2015-2016
Best Combined Physics 4H Student

SKILLS

PROGRAMMING
Languages:
Python • MatLAB
Familiar:
C • C++
Techniques:
Data Analysis • Bayesian Inference • Gaussian Process Regression • Cluster Computing • GIS analysis

LANGUAGES
French (Native language)
English (Fluent)
Italian (Basic to Intermediate)
Scottish Gaelic (Basic)

OTHER
Outreach • Teaching • Public Speaking • Graphic Design • Art